SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

Solution Summary

Field	Details
Repository Link / URL	
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	• HDI dataset preprocessing using Pandas and NumPy. • Machine Learning model for HDI category prediction. • Prediction of **Very High, High, Medium, and Low** HDI categories. • User-friendly Flask web application. • Model saving and loading using Joblib. • Data visualization using Matplotlib and Seaborn.
Pending / Incomplete Features	• User authentication and login. • Deployment on a cloud platform. • Integration with live HDI datasets. • Advanced analytics dashboard.
Setup / Run Instructions	**1.** Install Python 3.x and required packages (pip install -r requirements.txt). **2.** Download the HDI dataset (CSV) and place it in the project folder. **3.** Run model.py to train and save the machine learning model. **4.** Start the Flask application using python app.py. **5.** Open http://127.0.0.1:5000 in a web browser. **6.** Enter the required indicators and click **Predict** to view the HDI category.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- **The application follows a modular architecture with separate components for data preprocessing, model training, prediction, and the Flask web interface.**
- **The Machine Learning model is trained using Scikit-learn and saved using Joblib for efficient reuse.**
- **The system predicts the Human Development Index (HDI) category using Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita.**